GCSE Mathematics - Foundation Tier

Paper 3 (Calculator) Practice - Set 6

Time: 1 hour 30 minutes | Total marks: 80 | Calculator allowed

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Instructions

- Use black ink or ball-point pen.
- Answer ALL questions in the spaces provided.
- You must show all your working - answers given without working may not score full marks.
- Diagrams are NOT accurately drawn unless otherwise stated.
- If your calculator does not have a pi button, use pi = 3.142.

1.

Round 0.07346 to 2 significant figures.

(Total for Question 1 is 1 marks)

2.

Work out 2/3 of £216.

(Total for Question 2 is 2 marks)

3.

(a) Find the highest common factor (HCF) of 30 and 42. (1)

(b) Find the lowest common multiple (LCM) of 4 and 10. (1)

(Total for Question 3 is 2 marks)

4.

Simplify 8a - 3 - 5a + 11.

(Total for Question 4 is 2 marks)

5.

The stem-and-leaf diagram shows the test marks of 15 students. (Key: 2 | 5 means 25.)

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2 | 5  8
3 | 1  4  7  9
4 | 2  3  5  6  8
5 | 0  4  7  9
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(a) Write down the modal interval (group with the most marks). (1)

(b) Write down the median mark. (1)

(c) Work out the range of the marks. (1)

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(Total for Question 5 is 3 marks)

6.

Solve $2x + 5 = 17$.

(Total for Question 6 is 2 marks)

7.

A ladder is 5 m long. The base of the ladder is placed 1.4 m from a vertical wall.

[Diagram: right-angled triangle - wall (vertical), ground (horizontal), ladder (hypotenuse) 5 m, base 1.4 m from wall; height up the wall labelled h.]

Work out how high up the wall the top of the ladder reaches. Give your answer to 1 decimal place.

(Total for Question 7 is 3 marks)

8.

Here are the first four terms of an arithmetic sequence.

9, 14, 19, 24, ...

(a) Write down the next term. (1)

(b) Find an expression for the nth term of the sequence. (2)

(Total for Question 8 is 3 marks)

9.

The diagram shows two points A and B. North lines are drawn at both points. The bearing of B from A is shown as 075 deg.

[Diagram: point A with a north line; point B to the upper-right of A. Bearing of B from A = 075 deg.]

(a) Write down the bearing of B from A. (1)

(b) Work out the bearing of A from B. (2)

(Total for Question 9 is 3 marks)

10.

The formula for the area A of a parallelogram is $A = b \times h$.

(a) Work out A when b = 9 cm and h = 5 cm. (1)

The formula for the volume V of a cuboid is $V = l \times w \times h$.

(b) Work out V when l = 6 cm, w = 4 cm and h = 7 cm. (2)

(Total for Question 10 is 3 marks)

11.

A solid cube has a volume of 343 cm^3.

(a) Work out the length of one side of the cube. (2)

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(b) Hence work out the total surface area of the cube. (1)

(Total for Question 11 is 3 marks)

12.

Mia invests £3500 in an account paying 4% compound interest per year.

Work out the total value of her investment after 3 years. Give your answer to the nearest penny.

(Total for Question 12 is 4 marks)

13.

A bag contains 6 red counters and 4 blue counters. A counter is drawn at random, its colour noted, and then it is replaced. A second counter is then drawn.

(a) Write down the probability that the first counter drawn is red. (1)

(b) Work out the probability that BOTH counters drawn are red. (3)

(Total for Question 13 is 4 marks)

14.

A flagpole is held vertical by a straight support cable attached at the top of the pole and anchored to the ground 12 m from its base. The flagpole is 9 m tall.

[Diagram: right-angled triangle - vertical flagpole of height 9 m; ground distance 12 m; cable (hypotenuse) labelled c.]

(a) Show that the length of the cable is 15 m. (3)

A second cable, anchored 5 m from the base, runs to the top of the pole.

(b) Work out the length of this second cable. Give your answer to 1 decimal place. (2)

(Total for Question 14 is 5 marks)

15.

A closed cylindrical can has a radius of 3 cm and a height of 11 cm.

[Diagram: closed cylinder of radius 3 cm and height 11 cm.]

(a) Work out the volume of the can. Use pi = 3.142. Give your answer to 1 decimal place. (2)

(b) Work out the total surface area of the can. Give your answer to 1 decimal place. (3)

(Total for Question 15 is 5 marks)

16.

Here is a sequence:

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5,    12,    19,    26,    ...
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(a) Find an expression for the nth term. (2)

(b) Show that 100 is NOT a term of the sequence. (2)

(Total for Question 16 is 4 marks)

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17.

A car salesperson recorded the age and price of 9 used cars.

Age (yr)	1	2	3	4	5	6	7	8	10
Price (£)	15800	13200	11500	9700	8400	6900	5800	4600	3200

(a) On the grid provided, plot these data as a scatter graph. (2)

(b) Describe the correlation. (1)

(c) Draw a line of best fit. (1)

(d) Use your line of best fit to estimate the price of a 9 year old car. (1)

(Total for Question 17 is 5 marks)

18.

A swimming pool has the shape of a cuboid: 20 m long, 8 m wide and 1.5 m deep.

(a) Work out the volume of water needed to completely fill the pool. Give your answer in cubic metres. (2)

(b) Given that 1 m^3 = 1000 litres, work out the volume in litres. (1)

(c) A hose fills the pool at a rate of 250 litres per minute. Work out how long it takes to fill the pool. Give your answer in hours and minutes. (2)

(Total for Question 18 is 5 marks)

19.

On the map, the bearing of a church C from a school S is 130 deg. The actual distance from S to C is 4.5 km. The map uses a scale of 1 : 50 000.

(a) Work out the distance from S to C on the map. Give your answer in centimetres. (3)

(b) Write down the bearing of S from C. (2)

(Total for Question 19 is 5 marks)

20.

The first four terms of a sequence are 3, 7, 11, 15.

Let the nth term be T_n.

(a) Show that T_n = 4n - 1. (2)

(b) Work out T_25. (1)

(c) Show that the sum of the first two terms equals one-fifth of T_15. (2)

(Total for Question 20 is 5 marks)

21.

Tom is choosing between two savings accounts.

Account A: 3.5% compound interest per year.

Account B: £150 simple interest each year on a £2500 investment.

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Tom invests £2500 for 4 years.

(a) Work out the total in Account A after 4 years. Give your answer to the nearest penny. (3)

(b) Work out the total in Account B after 4 years. (2)

(c) Which account would you advise Tom to choose? Give a reason. (1)

(Total for Question 21 is 6 marks)

22.

A youth club has 80 members. The two-way table shows the type of activity members took part in last week, by age group.

	Sport	Music	Drama	Total
Under 14	18	9	___	35
14 plus	___	15	12	___
Total	___	___	20	80

(a) Complete the two-way table. (3)

(b) A member is chosen at random. Work out the probability that the member is aged 14 or over AND took part in Sport. (2)

(Total for Question 22 is 5 marks)

TOTAL FOR PAPER: 80 MARKS

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Mark Scheme

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Question 1

- 0.073 - B1

Total: 1 marks

Question 2

- 216 / 3 (= 72) - M1
- £144 - A1

Total: 2 marks

Question 3

- (a) 6 - B1
- (b) 20 - B1

Total: 2 marks

Question 4

- Either 3a or +8 collected - M1
- 3a + 8 - A1

Total: 2 marks

Question 5

- (a) 40 - 49 (or "40s") - B1
- (b) 43 (8th value) - B1
- (c) 59 - 25 = 34 - B1

Total: 3 marks

Question 6

- 2x = 12 - M1
- x = 6 - A1

Total: 2 marks

Question 7

- h^2 = 5^2 - 1.4^2 (Pythagoras for shorter side) - M1
- 25 - 1.96 = 23.04 - M1
- 4.8 (m) to 1 dp (h = sqrt(23.04) = 4.8) - A1

Total: 3 marks

Question 8

- (a) 29 - B1
- (b) Common difference 5 noted - M1
- (b) 5n + 4 - A1

Total: 3 marks

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Question 9

- (a) 075 (deg) - B1
- (b) 75 + 180 = 255 attempted - M1
- (b) 255 (deg) - A1

Total: 3 marks

Question 10

- (a) 9 x 5 = 45 (cm^2) - B1
- (b) 6 x 4 x 7 attempted - M1
- (b) 168 (cm^3) - A1

Total: 3 marks

Question 11

- (a) Cube root: 343 = 7^3 - M1
- (a) 7 (cm) - A1
- (b) 6 x 7^2 = 294 (cm^2) - B1

Total: 3 marks

Question 12

- Multiplier 1.04 seen - M1
- 3500 x 1.04^2 (= 3785.60) seen - M1
- 3500 x 1.04^3 - M1
- £3937.02 (accept £3937.024) - A1

Total: 4 marks

Question 13

- (a) 6/10 (or 3/5) - B1
- (b) Multiplication of independent events - M1
- (b) 6/10 x 6/10 (or 3/5 x 3/5) - M1
- (b) 36/100 (or 9/25 or 0.36) - A1

Total: 4 marks

Question 14

- (a) c^2 = 9^2 + 12^2 attempted - M1
- (a) 81 + 144 = 225 - M1
- (a) sqrt(225) = 15 (m) shown clearly - C1
- (b) c^2 = 9^2 + 5^2 = 81 + 25 = 106 - M1
- (b) sqrt(106) = 10.3 (m) - A1

Total: 5 marks

Question 15

- (a) V = pi x 3^2 x 11 (= 3.142 x 9 x 11) - M1
- (a) 311.1 (cm^3) accept 310.9 - 311.2 - A1
- (b) Two circular ends 2 x pi x 3^2 = 56.556 - M1
- (b) Curved surface 2 x pi x 3 x 11 = 207.372 - M1
- (b) 263.9 (cm^2) accept 263.8 - 264.0 - A1

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Total: 5 marks

Question 16

- (a) Common difference 7 noted; zeroth term -2 - M1
- (a) 7n - 2 - A1
- (b) 7n - 2 = 100 -> 7n = 102 -> n = 14.57... (not an integer) - M1
- (b) States 100 is NOT in the sequence with reason - A1

Total: 4 marks

Question 17

- (a) All 9 points plotted correctly - B2 (B1 for 6+ correct)
- (b) Negative correlation (strong) - B1
- (c) Sensible line of best fit - B1
- (d) Price at 9 yr (accept £3500 - £4200) - B1

Total: 5 marks

Question 18

- (a) 20 x 8 x 1.5 - M1
- (a) 240 (m^3) - A1
- (b) 240 x 1000 = 240 000 (litres) - B1
- (c) 240000 / 250 = 960 minutes; / 60 = 16 hours - M1
- (c) 16 hours 0 minutes - A1

Total: 5 marks

Question 19

- (a) 4.5 km = 450 000 cm - M1
- (a) 450000 / 50000 - M1
- (a) 9 (cm) - A1
- (b) 130 + 180 = 310 attempted - M1
- (b) 310 (deg) - A1

Total: 5 marks

Question 20

- (a) Common difference 4 noted, zeroth term -1 - M1
- (a) T_n = 4n - 1 shown clearly with calculation - C1
- (b) T_25 = 4(25) - 1 = 99 - B1
- (c) Sum of first two = 3 + 7 = 10; T_15 = 4(15) - 1 = 59; 59/5 = 11.8 (NOT 10) so the statement is FALSE - M1
- (c) Conclusion stated clearly (one-fifth of T_15 is 11.8, but sum of first two is 10, so statement is incorrect) - A1

Total: 5 marks

Question 21

- (a) Multiplier 1.035 seen - M1
- (a) 2500 x 1.035 repeated (or 1.035^4) - M1
- (a) £2868.81 (accept £2868.80 - £2868.82) - A1

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- (b) 150 x 4 = 600 - M1
- (b) 2500 + 600 = £3100 - A1
- (c) Account B is better (£3100 vs £2868.81) with reason - B1

Total: 6 marks

Question 22

- (a) Under 14 Drama = 35 - 18 - 9 = 8 - M1
- (a) Drama total 20 = 8 + 12 (consistent); 14+ Music = 15 known so 14+ Sport = ? - M1
- (a) 14+ total = 80 - 35 = 45; 14+ Sport = 45 - 15 - 12 = 18; column totals: Sport 36, Music 24, Drama 20 - A1
- (b) 14+ AND Sport = 18 out of 80 - M1
- (b) 18/80 = 9/40 - A1

Total: 5 marks

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Worked Solutions

Foundation Tier - Paper 3 (Calculator) - Set 6 | Detailed explanations

Question 1

Significant figures count from the first non-zero digit.

In 0.07346, the first non-zero digit is 7 (the 1st sf). The next digit is 3 (the 2nd sf).

Look at the next digit to decide rounding: 4 (< 5), so round down.

Answer: 0.073.

Question 2

2/3 of 216 means divide by 3 then multiply by 2.

216 / 3 = 72.

72 x 2 = 144.

Answer: £144.

Question 3

(a) Factors of 30: 1, 2, 3, 5, 6, 10, 15, 30. Factors of 42: 1, 2, 3, 6, 7, 14, 21, 42. HCF = 6.

(b) Multiples of 4: 4, 8, 12, 16, 20, 24, ... Multiples of 10: 10, 20, 30, ... First common = 20.

Question 4

a-terms: 8a - 5a = 3a.

Constants: -3 + 11 = 8.

Answer: 3a + 8.

Question 5

(a) Count leaves per row. Row 4 (40s) has 5 leaves - the most. Modal interval: 40 - 49.

(b) 15 students, so the median is the 8th value. Counting through: 2 + 4 + 5 puts the 11th in the 40s; the 8th value is in the 40s. Row 4 in order: 42, 43, 45, 46, 48. The 8th value overall is 43.

(c) Range = largest - smallest = 59 - 25 = 34.

Question 6

2x + 5 = 17. Subtract 5: 2x = 12. Divide by 2: x = 6.

Question 7

Pythagoras with the ladder as hypotenuse.

h^2 = 5^2 - 1.4^2 = 25 - 1.96 = 23.04.

h = sqrt(23.04) = 4.8.

Answer: 4.8 m to 1 dp.

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Question 8

(a) Common difference is +5. Next term = 24 + 5 = 29.

(b) Zeroth term = 9 - 5 = 4. nth term = 5n + 4.

Question 9

(a) Given: bearing of B from A = 075 deg.

(b) Going back from B to A, the bearing changes by 180 deg.

75 + 180 = 255 deg.

Answer: 255 deg.

Question 10

(a) A = b x h = 9 x 5 = 45 cm^2.

(b) V = 6 x 4 x 7 = 24 x 7 = 168 cm^3.

Question 11

(a) Volume of a cube = side^3, so side = cube root of 343.

7 x 7 x 7 = 343, so side = 7 cm.

(b) Surface area of a cube = 6 x side^2 = 6 x 49 = 294 cm^2.

Question 12

Multiplier = 1.04.

Year 1: 3500 x 1.04 = £3640.00.

Year 2: 3640 x 1.04 = £3785.60.

Year 3: 3785.60 x 1.04 = £3937.024.

To the nearest penny: £3937.02.

Question 13

(a) 6 red out of 10 counters: P(red) = 6/10 = 3/5.

(b) After replacement, the second draw is independent.

P(both red) = P(red) x P(red) = (6/10) x (6/10) = 36/100 = 9/25.

Question 14

(a) Right-angled triangle. Cable is the hypotenuse.

c^2 = 9^2 + 12^2 = 81 + 144 = 225.

c = sqrt(225) = 15 m. (Shown.)

(b) Same flagpole height 9 m, ground distance 5 m.

c^2 = 9^2 + 5^2 = 81 + 25 = 106.

c = sqrt(106) = 10.295..., i.e. 10.3 m.

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Question 15

(a) $V = \pi \times r^2 \times h = 3.142 \times 9 \times 11 = 311.058$, i.e. 311.1 cm^3.

(b) Surface area = 2 (circular ends) + curved side.

$2 \times \pi \times 3^2 = 2 \times 3.142 \times 9 = 56.556$.

$2 \times \pi \times 3 \times 11 = 207.372$.

Total = 56.556 + 207.372 = 263.928, i.e. 263.9 cm^2.

Question 16

(a) Common difference is +7. Zeroth term = 5 - 7 = -2. nth term = 7n - 2.

(b) For 100 to be in the sequence we need 7n - 2 = 100, i.e. 7n = 102, so n = 102 / 7 = 14.571...

n is not a whole number, so 100 is NOT a term of the sequence.

Question 17

(a) Plot 9 points.

(b) Negative correlation - older cars cost less.

(c) Best fit drawn.

(d) At 9 years the line gives roughly £3500 - £4200.

Question 18

(a) Volume = $20 \times 8 \times 1.5 = 240$ m^3.

(b) 240 x 1000 = 240 000 litres.

(c) Time = volume / rate = 240 000 / 250 = 960 minutes.

960 / 60 = 16 hours exactly.

Answer: 16 hours 0 minutes.

Question 19

(a) 4.5 km = 4500 m = 450 000 cm.

Map distance = 450 000 / 50 000 = 9 cm.

(b) Reverse bearing = 130 + 180 = 310 deg.

Question 20

(a) Differences: 7-3=4, 11-7=4, 15-11=4. Common difference 4.

Zeroth term = 3 - 4 = -1. So $T_n = 4n - 1$. (Shown.)

(b) $T_{25} = 4(25) - 1 = 100 - 1 = 99$.

(c) Sum of first two = 3 + 7 = 10.

$T_{15} = 4(15) - 1 = 60 - 1 = 59$.

One-fifth of $T_{15} = 59 / 5 = 11.8$.

But the sum of the first two is 10, NOT 11.8 - so the stated equality is FALSE.

(Worked solution shows the claim must be disproved as part of the show that.)

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Question 21

(a) Account A: 2500 x 1.035^4 = 2500 x 1.147523... = £2868.81 (to nearest penny).

(b) Account B: simple interest £150 per year for 4 years = 4 x 150 = £600.

Total in B = 2500 + 600 = £3100.

(c) £3100 > £2868.81, so Account B gives a larger return. Advise Account B.

Question 22

(a) Under 14 Drama = 35 - 18 - 9 = 8.

Total members 14+: 80 - 35 = 45.

14+ Sport = 45 - 15 - 12 = 18.

Column totals: Sport 18+18=36, Music 9+15=24, Drama 8+12=20. Check: 36+24+20=80.

(b) P(14+ AND Sport) = 18 / 80 = 9/40.

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